ABHITAY SHINDE

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Data scientist who builds customer-facing GenAI systems end to end: agentic retrieval, evaluation, and guardrails. Backs each launch with controlled experiments and causal designs, so the impact claim holds up as well as the model does.

EDUCATION

Columbia University Sep 2024 - Dec 2025
M.S. Data Science (AI Specialization), GPA: 3.7/4.0 New York, NY

PROFESSIONAL EXPERIENCE

Ask2AI New York, NY
Data Scientist, Customer Acquisition *Feb 2026 - Present*

- Shifted 22% of acquisition budget off channels that were absorbing credit for organic conversions, by replacing last-touch reporting with difference-in-differences designs and uplift XGBoost models that isolated incremental campaign impact.
- Unified 11 source systems into a single 30M-customer modeling dataset, by building a Python and Polars feature and causal inference pipeline, correcting a selection bias that had inflated the prior scoring model's estimated lift.
- Lifted customer LTV by 12% and cut churn by 25%, by pairing uplift modeling with MILP to reallocate budget across channels.

Julius Baer New York, NY
Data Scientist, Generative AI and Personalization (Contract) *Jun 2025 - Sep 2025*

- Cut personalized client document drafting from 4 hours to 30 minutes, by building a LangGraph agent that pulled holdings, preferences, and market news into a single draft.
- Measured a 25% lift in engagement, by comparing recipient open and read-through rates for drafts against controls in SQL.
- Automated compliance review at 86% agreement with human reviewers, by scoring every draft against the firm's rubric with an LLM judge and regenerating failures.
- Cut relationship manager's selection time by 35%, by shipping the pipeline as a chat interface where they revise drafts in place.

TD Bank New York, NY
Data Scientist Capstone, Customer Analytics and Segmentation *Jan 2025 - May 2025*

- Raised fraud model precision-recall AUC from 0.20 to 0.67 on 100 million transactions, and quantified a 14% incremental catch rate over the incumbent rules engine that leadership used to approve deployment.
- Defined 7 behavioral segments from transaction patterns, and automated risk-tier reporting that replaced a manual monthly review.

SELECTED PROJECTS

Product Support Assistant over Appliance Manuals | LangGraph, Neo4j, FAISS, LLM-as-Judge (Link)

- Answered repair questions that required chaining facts across separate manual sections, raising accuracy from 78% to 84% on a 100-question test set, by breaking each question into sub-queries and following part-to-symptom links in a knowledge graph.
- Surfaced the correct manual section 30% more often than semantic search alone, by matching exact error codes with keyword search and vague symptom descriptions with vector search, then grading both approaches with an LLM judge.

Semantic Cache for LLM Serving | Semantic Caching, Embeddings, FAISS, Python (Link)

- Cut LLM API calls by 72% and median response time by 3.5x across 5,000 support questions, by recognizing when a new question means the same thing as one already answered and reusing the stored answer.
- Held wrong answers from cache hits to 1.35%, under a 2% ceiling, by tuning the similarity cutoff against 1,945 question pairs that humans had labeled as duplicates or not.

Fine-Tuned, Guarded Text-to-SQL Copilot | LoRA, MLX, sqlglot, Guardrails, Python (Link)

- Improved SQL generation accuracy by 12% relative (42.5% to 47.5% on the Spider benchmark) training entirely on a laptop, by fine-tuning a quantized 7B model at 1.1 GB peak memory with no cloud GPU spend.
- Blocked all 300 attacks in a red-team test across 20 databases, by inspecting the structure of every generated query and permitting only read-only lookups against approved tables.

TECHNICAL SKILLS

- **LLM Systems**: Agentic AI (LangGraph), RAG (Hybrid, Graph RAG), Tool & Function Calling, Prompt Engineering, Structured Outputs, Fine-Tuning (LoRA/QLoRA), Guardrails, Human-in-the-Loop
- **Evaluation & Reliability**: LLM-as-a-Judge, Eval Set Design, Regression Testing, Hallucination Mitigation, Red Teaming
- **Serving & Cost**: Semantic Caching, Latency Optimization, Embeddings, Vector DBs (FAISS, ChromaDB), Neo4j
- **Customer Measurement**: A/B Testing, Causal Inference (DiD, PSM, Double ML), Uplift Modeling, Metric Design, Personalization, Segmentation, LTV & Churn Modeling, MMM
- **Engineering**: Python, SQL, PySpark, Polars, Pandas, PyTorch, XGBoost, Airflow, Docker, Git, CI/CD, AWS, GCP, Databricks